

ConstantPad1d#

<https://pytorch.org/docs/stable/generated/torch.nn.ConstantPad1d.html>

Description:

Pads the input tensor boundaries with a constant value. For N-dimensional padding, use `torch.nn.functional.pad()`. `padding` (int, tuple) – the size of the padding. If is int, uses the same padding in both boundaries. If a 2-tuple, uses `(padding_left\text{padding\_left}padding_left, padding_right\text{padding\_right}padding_right)` Input: `(C,Win)(C, W_{in})` `(C,Win)` or `(N,C,Win)(N, C, W_{in})(N,C,Win)`. Output: `(C,Wout)(C, W_{out})` `(C,Wout)` or `(N,C,Wout)(N, C, W_{out})(N,C,Wout)`, where

Function Signature

```
class torch.nn.ConstantPad1d(padding, value)[source]#
```

Parameters

Shape:

Code Examples

```
>>> m = nn.ConstantPad1d(2, 3.5)
>>> input = torch.randn(1, 2, 4)
>>> input
tensor([[[[-1.0491, -0.7152, -0.0749,  0.8530],
          [-1.3287,  1.8966,  0.1466, -0.2771]]]])
>>> m(input)
tensor([[[[ 3.5000,  3.5000, -1.0491, -0.7152, -0.0749,  0.8530,
           3.5000,
            3.5000],
          [ 3.5000,  3.5000, -1.3287,  1.8966,  0.1466, -0.2771,
           3.5000,
            3.5000]]]])
>>> m = nn.ConstantPad1d(2, 3.5)
>>> input = torch.randn(1, 2, 3)
>>> input
tensor([[[[ 1.6616,  1.4523, -1.1255],
          [-3.6372,  0.1182, -1.8652]]]])
>>> m(input)
tensor([[[[ 3.5000,  3.5000,  1.6616,  1.4523, -1.1255,  3.5000,
           3.5000],
          [ 3.5000,  3.5000, -3.6372,  0.1182, -1.8652,  3.5000,
           3.5000]]]])
>>> # using different paddings for different sides
>>> m = nn.ConstantPad1d((3, 1), 3.5)
>>> m(input)
tensor([[[[ 3.5000,  3.5000,  3.5000,  1.6616,  1.4523, -1.1255,
           3.5000],
          [ 3.5000,  3.5000,  3.5000, -3.6372,  0.1182, -1.8652,
           3.5000]]]])
```

Detailed Documentation

Documentation

Pads the input tensor boundaries with a constant value.

For N-dimensional padding, use `torch.nn.functional.pad()`.

`padding` (int, tuple) – the size of the padding. If is int, uses the same padding in both boundaries. If a 2-tuple, uses $(padding_left, padding_right)$.

Input: (C, W_{in}) or (N, C, W_{in}) .

Output: (C, W_{out}) or (N, C, W_{out}) , where

$$W_{out} = W_{in} + padding_left + padding_right$$